18 (a) Show that the equation $x^3 + x = 7$ has a solution between 1 and 2

Let $f(x) = x^3 + x - 7$.

$f(1) = 1^3 + 1 - 7 = -5$

$f(2) = 2^3 + 2 - 7 = 3$

The polynomial is continuous and changes sign, so a root lies between 1 and 2.

1 mark
Substitute both endpoints correctly

1 mark
Explain why the sign change gives a solution

(2)

(b) Show that the equation $x^3 + x = 7$ can be rearranged to give $x = \sqrt[3]{7 - x}$

$x^3 + x = 7$

$-x$ $-x$

$x^3 = 7 - x$

Take the cube root: $x = \sqrt[3]{(7 - x)}$.

1 mark
Subtract x, then take the cube root on both sides

(1)

(c) Starting with $x_0 = 2$,
use the iteration formula $x_{n+1} = \sqrt[3]{7 - x_n}$ three times to find an estimate for a solution of $x^3 + x = 7$

Start $x_0 = 2$ and use the previous result each time.

$x_1 = \sqrt[3]{(7 - 2)} = 1.7099759467...$

1 mark
Substitute $x_0 = 2$

$x_2 = \sqrt[3]{(7 - 1.7099759467...)} = 1.7424188025...$

1 mark
Substitute the first iterate

$x_3 = \sqrt[3]{(7 - 1.7424188025...)} = 1.7388495056...$

Keep the calculator's full previous answer between iterations.

1 mark
Third iterate, correctly rounded

1.74

(3)

(Total for Question 18 is 6 marks)